

CS201 Midterm 3

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Abstract—Write only ONE function per question in Java. Do not check the input, assume that the input is correct. Unless noted, (i) the input data structure is not empty, (ii) time complexity of the algorithm does not matter, (iii) you can not use extra data structures, (iv) you can use any function given in the data structure.

I. QUESTION (HEAP), 20 POINTS

Write a method in **MinHeap** class

```
double averagePercolateUp(int[] list)
```

that returns the average number of percolate up operations for numbers given in the list, if each number is assumed to be inserted independently to the heap. Do not modify the heap while checking.

II. QUESTION (HEAP), 15 POINTS

Write the method in **MaxDHeap** class

```
int secondMinimum()
```

that finds the second minimum item. You may not use any additional data structures. You may assume there are more than 2 items in the heap. You should not use more than N comparisons (Sum of ifs else ifs else is at most N)

III. QUESTION (DISJOINT SET), 15 POINTS

Write the method in **DisjointSet** class

```
int sameHeightSets(int index)
```

which returns the number of sets that have same height as the set with given index. The height of an element is calculated by the number of nodes visited when we traverse the tree starting from that set and continue through the parent links until the top. Do not use any class or external methods.

IV. QUESTION (DISJOINT SET), 15 POINTS

Write the method in **DisjointSet** class

```
int childrenParentEqual()
```

which returns count of sets where direct children's (not including grandchildren) total data value is equal to set's parent's data value. Root node's parent is itself and a leaf node does not have any children. Do not use any class or external methods.

V. QUESTION (GRAPH), 15 POINTS

Write a new constructor for Graph class for linked list implementation

```
Graph(Graph[] graphs, int minCount)
```

which constructs a new graph by intersection of all graphs in the graph array, where edge is included in the constructed when such edge appears in at least minCount number of graphs. Size represents the number of graphs in graphs variable. You may assume that all graphs have the same size and same set of vertices.

VI. QUESTION (GRAPH), 20 POINTS

Write the method in array (adjacency matrix) implementation of undirected Graph

```
bool isTwoGraph()
```

which returns whether graph is a two-graph graph or not. You may assume graph is connected. A two-graph is a graph where degree of all vertices are 2. You are not allowed to use depth first search or breadth first search.